EXAMINATION NO.:_____

THE MALAWI NATIONAL EXAMINATION BOARD

2021 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

PHYSICS

Subject Number: M164/II

Friday, 08 January

Time Allowed: 2 hour sessions
10:00 am onwards

PAPER II

(40 marks)

Practical

Instructions

1. **This paper contain 6 printed pages. Please check.**
2. Fill in your **Examination Number** at the top of each page.
3. Answer **all** the **4** questions in the spaces provided.
4. Use of electronic calculator is allowed.
5. The maximum number of marks for each answer is indicated against each question.
6. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			

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Section A (20 marks)

1. With the aid of a diagram, describe an experiment that could be carried out to determine the required length of nichrome wire to be cut from a roll of the wire such that the cut piece of wire has a resistance of 2Ω.

(10 marks)

Continued/…

2. Design an experiment that could be carried out to determine the density of Irish potato.

(10 marks)

Continued/…

Section B (20 marks)

3. You are provided with a piece of thread, 100g bob, clamp stand and stop watch.

a) Set up the apparatus as shown in figure 1.

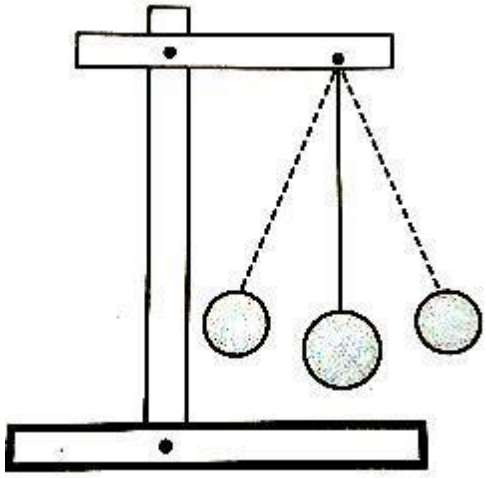


Figure 1

b) With the string measuring 50cm, displace the bob for an amplitude of 10cm as shown in figure 1 and allow it to swing freely.

c) Record the time taken for 10 vibrations in table 1.

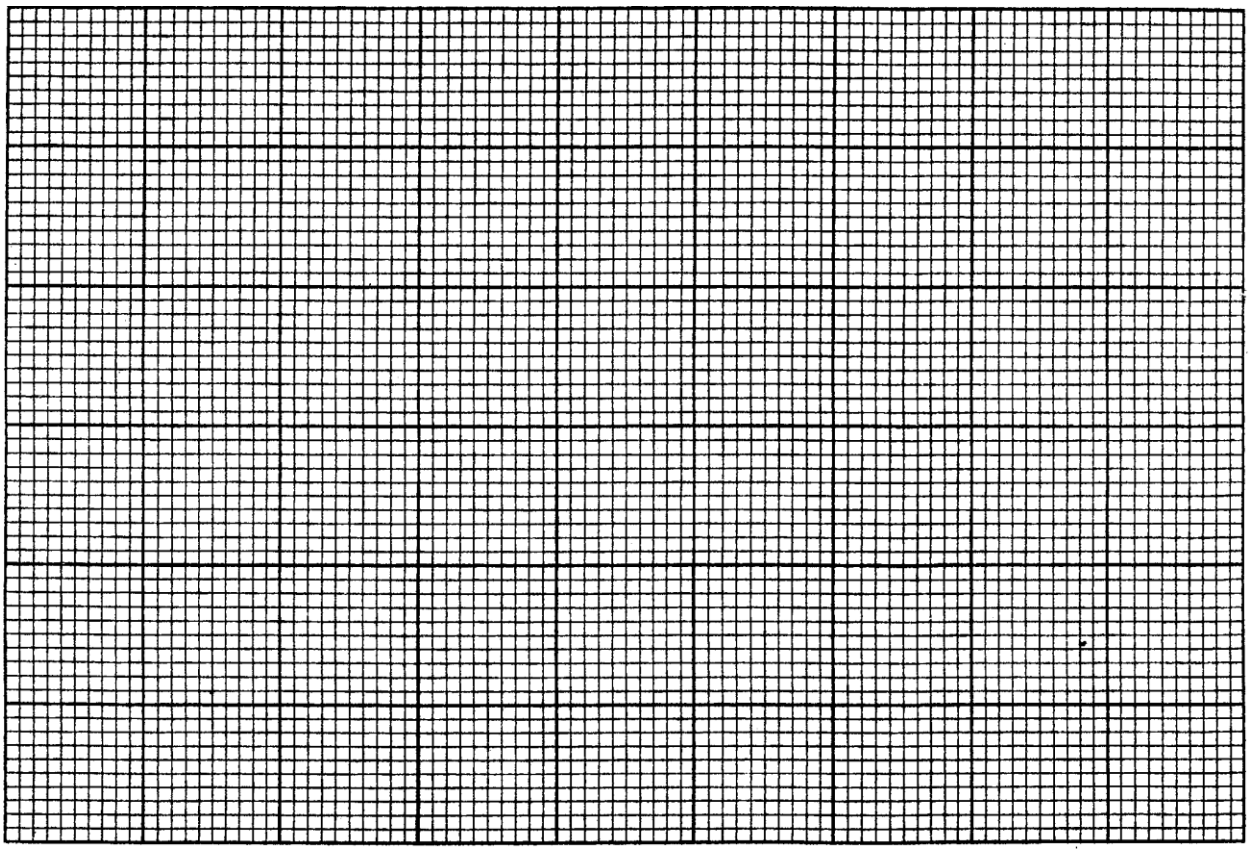
Length of string (cm)	Time Taken for 10 vibrations	Frequency (10 oscillations/time taken)
50		
40		
30		
20		
10		

Table 1

d) Repeat procedure b and c with strings of length 40, 30, 20 and 10cm.

e) Plot a graph on page 5 of frequency against length of strings.

Continued/…



f). **Use the graph to describe the relationship between frequency and length of strings.**

_____

_____

4. You are provided with a measuring cylinder, 2 ball bearings, a 30cm ruler, marker and water.

a) Pour water into the measuring cylinder to the 20cm^3 mark.

b) Mark the level of water as Y_0.

c) Draw a diagram of the cylinder to show the position of the water meniscus

Continued/…

d) Drop a ball bearing into the measuring cylinder and mark the new water level Y_1.

e) Measure the increase in height of water column. (Y_1-Y_0)

_____

f) Calculate the pressure of water at point Y_0. (density of water 1g/cm^3, gravity = 10m/s^2)

Pressure (1 pa + dgh) = _____

g) Drop the second ball bearing into the measuring cylinder and mark the new water level Y_2.

h) Measure the increase in height $Y_2 – Y_0$.

_____

i) Calculate pressure at point Y_0. Pressure = 1 pa + dgh.

_____

j) What conclusion can be drawn from the experiment?

_____

_____

END OF QUESTION PAPER

NB: This paper contains 6 printed pages.